MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 17 of 30 (Start time: 02:35:09 PM, 29 December 2020)

Total Marks: 1

If ϕ and ψ are scalar point functions with continuous second order derivatives in a region R bounded by a closed surface S then Green's first identity states that

Select the correct option

Reload Math Equations

- $\int\int\int_R(\phi\nabla^2\psi+\nabla\phi\nabla\psi)dV=\int\int_S(\phi\nabla\psi).d\vec{S}$
- $\int\int\int_R(\phi\nabla^2\psi+\nabla\phi\nabla\psi)dV=\int\int_S(\psi\nabla\phi)d\vec{S}$

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver=f937b7cf-e297-4109-afe4-aa39d0855144

MC190403973 LAILA SAIF

MTH622:Grand Quiz

Question # 29 of 30 (Start time: 11:48:44 AM, 29 December 2020)

The unit normal to the surface $z = 0$ is

Select the correct option

- $\pm \hat{j}$
- $\pm \hat{i}$
- $\pm \hat{k}$
- does not exist

Scanned with CamScanner

3:47 p.m.

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 10 of 30 (Start time: 03:47:45 PM, 29 December 2020)

Total Marks: 1

________ is the example of scalar field.

Select the correct option

- Velocity
- Displacement
- Acceleration
- Pressure

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MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 0

Question # 23 of 30 (Start time: 03:50:16 PM, 29 December 2020)

The unit vector $\hat{i}$ is normal to the surface

Select the correct option

○	$x = 0$
○	$y = 0$
○	$z = 0$
○	none of these

46% 3:44 p.m.

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 7 of 30 (Start time: 03:44:42 PM, 29 December 2020)

Total Marks: 1

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = \hat{k}$ becomes

Select the correct option

Reload Math Equations

○	$\oint_C (dy\hat{i} - dx\hat{j})$
○	$\oint_C (dz\hat{i} - dx\hat{k})$

Click to Save Answer & Move to Next Question

15:56 68%

Quiz

https://quiz.vu.edu.pk/QuizQuestio...

MC200205156: NIGHAT NOREEN

Time Left 77 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 3 of 30 (Start time: 03:56:02 PM, 29 December 2020)

Total Marks: 1

$\hat{i} \times \vec{\nabla}\varphi$ for $\varphi = x$ becomes

Select the correct option

Reload Math Equations

○	$\vec{0}$
○	$\hat{j}$

Click to Save Answer & Move to Next Question

Question # 5 of 30 (Start time: 10:59:17 AM, 29 December 2020)

Newton's first law is also called

Select the correct option

- law of inertia
- law of conservation of momentum

Type here to search

Question # 9 of 30 (Start time 11 03 08 AM, 29 December 2020)

The ______ is the rate at which the function f(x,y,z) changes at a point

(x_0, y_0, z_0)

in the direction

Select the correct option

○	gradient
○	curl
○	directional derivative
○	divergence

Type here to search

46% 3:49 p.m.

Quiz

MC200203833: FAIZAN HASSAN

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 12 of 30 (Start time: 03:49:36 PM, 29 December 2020)

Total Marks: 1

The unit normal to the surface $x = 0$ is

Select the correct option

Reload Math Equations

○	$\pm \hat{j}$
○	$\pm \hat{i}$
○	$\pm \hat{k}$
○	does not exist

190403973: LAILA SAIF

H622:Grand Quiz

Quiz Start Time: 11:07 AM, 29 December 2020

stion # 8 of 30 (Start time: 11:18:12 AM, 29 December 2020)

Total Marks: 1

een's theorem states that if R is simply connected region of the xy-plane bounded by a closed curve C and if M and N are continuous functions of x and y having continuous derivatives in R then

Reload Math Equations

ect the correct option

- $\oint_C Mdy + Ndx = \int\int_R(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y})dxdy$
- $\oint_C Mdx + Ndy = \int\int_R(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y})dxdy$

__/__/ 2020

$v = 2xi + azk$

$\nabla \cdot v = 0$

$\nabla \cdot (2x\hat{i} + az\hat{k}) = 0$

$\frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial z}(az) =$

$2 + a = 0$

$a = -2$

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quiz.vu.edu.pk/QuizQuestion.aspx?ver=cd0580e3-18ec-478f-98f9-...

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MC190406995: AKHTAR HASSAN NOORI

MTH622 Grand Quiz

Quiz Start Time: 03:28 PM

Question # 17 of 30 (Start time: 03 42 52 PM, 29 December 2020)

The identity $\oint_C \varphi d\vec{r} = -\int\int_S \vec{\nabla}\varphi \times d\vec{S}$ is

Select the correct option

Reload

○	true
○	false

Question # 3 of 30 (Start time: 12:00:07 PM, 29 December 2020)

A simple closed curve does not intersects itself anywhere.

Select the correct option

- true
- false

MC190401710: MALIK MUHAMMAD QASIM

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 20 of 30 (Start time: 02:38:31 PM, 29 December 2020)

Total Marks: 1

$\int_C d\vec{r} \times \vec{B}$ for $\vec{B} = y\hat{j}$ becomes

Select the correct option

Reload Math Equations

○	$\int_C (y\,dz\,\hat{i} + y\,dy\,\hat{k})$
○	$\int_C (-y\,dz\,\hat{i} + y\,dx\,\hat{k})$

MC100403340: ANSA SAID

MTH622:Grand Quiz

Question # 18 of 30 (Start time: 11:11:45 AM, 29 December 2020)

Gradient is a vector that represents both the magnitude and the direction of the maximum space rate of increase of a ___

Select the correct option

- ○ scalar
- ○ vector
- ○ directional derivative
- ○ divergence

Type here to search

MC190203990: MUHAMMAD REHAN MURAD

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 25 of 30 (Start time: 03:23:34 PM, 29 December 2020)

Total Marks: 1

Evaluate $\int_C$ (e^[illegible]) dy along the line segment joining $(0, 0, 0)$ to $(0, 1, 0)$.

Select the correct option

Reload Math Equations

○	1
○	-1

Click to Save Answer & Move to Next Question

MC190203990: MUHAMMAD REHAN MURAD

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 10 of 30 (Start time: 03:04:00 PM, 29 December 2020)

Total Marks:

The radial component of velocity for a particle moving in circular path is

Select the correct option

- constant
- radius itself
- variable
- zero

https://quiz.vu.edu.pk/QuizQuestion.asp

Evaluate $\int_c (x+y)dy$ along the line segment joining $(0,0)$ to $(0,1)$

Reload Math E

Select the correct option

- ○ $\frac{1}{3}$
- ○ $\frac{1}{2}$

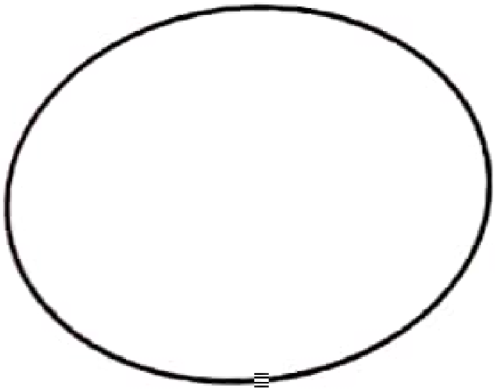
Module No. 67

Simply and Multiply Connected Re

Simply Connected Region

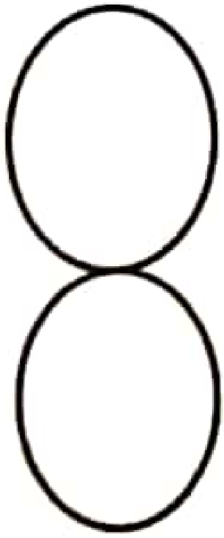
A simple closed curve is a closed curve which does not intersects itself anywher

the curve in the figure (i) is a simple closed curve



(i)

while the curve the curve in figure (ii) is not a simple closed curve.



(ii)

A region R which is said ...

 Total Marks: 1

Newton's first law is also called

Select the correct option

- law of inertia
- law of conservation of momentum

Click to Save Answer & Move to Next Question

MC190403548: ANSA SAID

MTH622:Grand Quiz

Question # 22 of 30 (Start time: 11:16:46 AM, 29 December 2020)

The unit normal to the surface $z = 0$ is

Select the correct option

- $\pm\hat{j}$
- $\pm\hat{i}$
- $\pm\hat{k}$
- does not exist

Type here to search

MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 12 of 30 (Start time: 03:07:22 PM, 29 December 2020)

Total Marks: 1

If $\hat{n}$ is the unit normal to the surface S and ψ is a scalar point function having continuous partial derivatives then $\vec{\nabla}\psi.\hat{n} = \frac{\partial\psi}{\partial n}$ implies

Select the correct option

Reload Math Equations

○	$\vec{\nabla}\psi.d\vec{S} = \frac{\partial\psi}{\partial n}dS$
○	$\vec{\nabla}\psi.d\vec{S} = \frac{\partial\psi}{\partial n}\hat{n}$

Click to Save Answer & Move to Next Question

3:59 p.m. 44%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 20 of 30 (Start time: 03:59:50 PM, 29 December 2020)

Total Marks: 1

$\int_C d\vec{r} \times \vec{B}$ for $\vec{B} = x\hat{i}$ becomes

Select the correct option

Reload Math Equations

O	$\int_C (xdz\hat{j} - xdy\hat{k})$
O	$\int_C (xdz\hat{i} + xdy\hat{k})$

vergance is vector of scalar

, All Images News Videos More

bout 5,120,000 results (0.46 seconds)

;howing results for ***divergence*** is vector ***or*** scalar
;earch instead for divergance is vector of scalar

The **divergence** of a **vector** field simply measures how much the flow is expanding at a given point. It does not indicate in which direction the expansion is occuring. Her (in contrast to the curl of a **vector** field), the **divergence** i a **scalar**.

mathinsight.org › divergence_idea

The idea of the divergence of a vector field - M

Abo

People also ask

How do you find the divergence of a vector?

Jazz 17.9K/s 4G 37% 2:43 PM

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MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 26 of 30 (Start time: 02:43:14 PM, 29 December 2020)

Total Marks: 1

The property $\nabla.\vec{V} = \vec{V}.\nabla$ always holds.

Select the correct option

Reload Math Equations

○	True
○	False

Click to Save Answer & Move to Next Question

MC200203833: FAIZAN HASSAN

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 17 of 30 (Start time: 03:55:42 PM, 29 December 2020)

Total Marks: 1

According to Stokes' theorem if S is an open, two sided surface bounded by a simple closed curve C, and A has continuous first partial derivatives then

$\oint_C A.dr = ?$

Select the correct option

Reload Math Equations

○	$\iint_S (\nabla . A) dS$
○	$\iint_S (\nabla \times A) dS$
○	$\iint_S (\nabla \times A) . \hat{n} dS$
○	$\iint_S (\nabla \times \hat{n}) . A dS$

Click to Save Answer & Move to Next Question

Quiz
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MC190203990: MUHAMMAD REHAN MURAD

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 202

Question # 16 of 30 (Start time: 03:11:43 PM, 29 December 2020)

Total Marks:

The Stoke's theorem can be used to find which of the following?

Select the correct option

- Surface area enclosed by a function in the given region
- Volume enclosed by a function in the given region
- Linear distance
- Curl of the function

4:02 p.m. 44%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 23 of 30 (Start time: 04:02:57 PM, 29 December 2020)

Total Marks: 1

The Stoke's theorem can be used to find which of the following?

Select the correct option

- Surface area enclosed by a function in the given region
- Volume enclosed by a function in the given region
- Linear distance
- Curl of the function

perator which is obtained by taking the divergence

is denoted as

$$\nabla . \nabla = \nabla^2 = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right.$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

It is a scalar operator.

In one and two dimension, the Laplace operator reduce

$$\nabla^2 = \frac{\partial^2}{\partial x^2}$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

If $\varphi(x, y, z)$ is a scalar point function, then the divergenc

$\nabla^2\varphi$ is called the Laplacian of φ and the equation $\nabla^2\varphi =$

If a scalar function φ satisfies the Laplace equation $\nabla^2\varphi =$

said to be a harmonic function in the region R

ch

Which of the following theorem convert line integral to surface integral?

Select the correct option

- Stoke's theorem only
- Gauss divergence and Stoke's theorem
- Green's theorem only
- Stoke's and Green's theorem

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622 Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 28 of 30 (Start time: 02:44:53 PM, 29 December 2020)

Total Marks: 1

The parametric equations of a circle of radius 5 are

Select the correct option

Reload Math Equations

- $x = \sqrt{5}\cos t$, $y = \sqrt{5}\sin t$
- $x = 25\cos t$, $y = 25\sin t$
- $x = 5 sint$, $y = 5\sin t$
- $x = 5\cos t$, $y = 5\sin t$

Click to Save Answer & Move to Next Question

MC200205156: NIGHAT NOREEN

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 8 of 30 (Start time: 04:00:05 PM, 29 December 2020)

Total Marks: 1

Vector is a quantity which is specified by ______ and _______.

Select the correct option

- scalar function, magnitude
- scalar function, direction
- magnitude, direction
- All of above

Click to Save Answer & Move to Next Question

MC190403973: LAILA SAIF

MTH622:Grand Quiz

Question # 7 of 30 (Start time: 11:17:00 AM, 29 December 2020)

Quiz Start Ti

If $\vec{T}$ is a unit vector to a space curve C and S is the arc length then $\frac{d\vec{T}}{dS}$ is

Select the correct option

- parallel to $\vec{T}$
- perpendicular to $\vec{T}$

Question # 7 of 30 (Start time: 02:31:12 PM, 28 December 2020)

$\nabla \cdot \vec{V}$ is a ________ quantity

Select the correct option

○	Scalar
○	Vector

Type here to search

MC190403973: LAILA SAIF

MTH622:Grand Quiz

Quiz Start Time

Question # 6 of 30 (Start time: 11:15:18 AM, 29 December 2020)

A vector $\vec{A}$ is called ir-rotational if ______.

Select the correct option

- curl $\vec{A} \neq 0$
- gard $\vec{A} = 0$
- grad $\vec{A} \neq 0$
- curl $\vec{A} = 0$

Question # 6 of 30 (Start time: 12:05:10 PM, 29 December 2020) Total Marks: 1

The Laplace operator ∇^2 is a ________ operator

Select the correct option

Reload Math Equations

○	Scalar
○	Vector

Click to Save Answer & Move to Next Question

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MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 22 of 30 (Start time: 02:40:45 PM, 29 December 2020) **Total Marks: 1**

A particle moves in a plane in such a way that at any time t its distance from the origin is $r = 2t + 3t^2$ and its angle with positive x-axis is $\theta = \sqrt{t}$. Find the radial and transverse components of velocity.

Select the correct option

Reload Math Equations

○	$2 + 6t$ and $(\sqrt{t} + \frac{3}{2}t^{\frac{3}{2}})$
○	$2 + 6t$ and $\frac{1}{2\sqrt{t}}$

Click to Save Answer & Move to Next Question

Question # 4 of 30 (Start time: 12:01:32 PM, 29 December 2020)

Which of the following objects has more inertia?

Select the correct option

- an object with mass $40kg$
- an object with mass $30kg$

Divergence is ______ operator.

Select the correct option

○	scalar
○	vector
○	directional
○	None of these

3:25 pm 53%

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MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 26 of 30 (Start time: 03:25:17 PM, 29 December 2020)

Total Marks: 1

If $\varphi = xy$, then $\nabla\varphi$ is

Select the correct option

Reload Math Equations

○	$y\hat{i} + x\hat{j}$
○	$x\hat{i} + y\hat{j}$
○	$y\hat{i} - x\hat{j}$
○	$y\hat{j}$

Click to Save Answer & Move to Next Question

3:19 pm 55%

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MC190203990: MUHAMMAD REHAN MURAD

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 23 of 30 (Start time: 03:19:22 PM, 29 December 2020) Total Marks: 1

The Laplacian of ϕ is also called as

Select the correct option

Reload Math Equations

○	Gradient of divergence of ϕ
○	Divergence of gradient of ϕ
○	Curl of ϕ
○	Divergence of curl of ϕ

Click to Save Answer & Move to Next Question

2:57 pm 62%

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MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 5 of 30 (Start time: 02:57:16 PM, 29 December 2020)

Total Marks: 1

If a scalar function φ satisfies the Laplace equation

$$\nabla^2\varphi = 0$$

In a Cartesian region R, then φ is said to be a _____ function in the region R.

Select the correct option

Reload Math Equations

- borel
- harmonic
- baire
- lipschitz

Click to Save Answer & Move to Next Question

 Total Marks: 1

The infinitesimal rotation of a vector field is described by the ________.

Select the correct option

○	Divergence
○	Gradient
○	Curl
○	Scalar

Click to Save Answer & Move to Next Question

46% 3:47 p.m.

Quiz

MC200203833: FAIZAN HASSAN

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 9 of 30 (Start time: 03:47:08 PM, 29 December 2020)

Total Marks: 1

Evaluate $\int_0^1 \int_0^1 \int_0^1 y \, dx dy dz =$

Select the correct option

Reload Math Equations

○	$\frac{1}{2}$
○	$\frac{-1}{2}$

QuizQuestion.aspx?ver=b6ff9200-3151-4721-ba10-a220f805ca85

MC190403973 LAILA SAIF

MTH622:Grand Quiz

Question # 30 of 30 (Start time: 11:50:14 AM, 29 December 2020)

The Stoke's theorem uses which of the following operation?

Select the correct option

- Divergence
- Gradient
- Curl
- Laplacian

ZONG
Jazz

3G 4G 79% 10:59 AM

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31

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 10:43 AM, 29 December 202

Question # 17 of 30 (Start time: 10:59:24 AM, 29 December 2020)

Total Marks:

According to first law of motion if some external force is acting on a body then the velocity of object is constant.

Select the correct option

- true
- false

Click to Save Answer & Move to Next Question

Question # 17 of 30 (Start time: 11:11:07 AM, 29 December 2020)

$\int_C K\vec{A}.d\vec{r}$=_________.

Select the correct option

○	$K\int_C \vec{A}.d\vec{r}$
○	$\vec{A}\int_C K.d\vec{r}$
○	$\int_C \vec{A}(K.d\vec{r})$
○	$\int_C (\vec{A}.d\vec{r})K$

Type here to search

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 1 of 30 (Start time: 02:22:58 PM, 29 December 2020)

Total Marks: 1

If φ and ψ are scalar point functions then $\nabla\left(\frac{\varphi}{\psi}\right) =$

Select the correct option

Reload Math Equations

- $\frac{\psi\nabla\varphi - \varphi\nabla\psi}{\psi^2},\ \psi = 0$
- $\frac{\psi\nabla\varphi + \varphi\nabla\psi}{\psi^2},\ \varphi = 0$
- $\frac{\psi\nabla\varphi - \varphi\nabla\psi}{\psi^2},\ \psi \neq 0$
- $\frac{\psi\nabla\varphi - \varphi\nabla\psi}{\psi^2},\ \psi = 0$

Click to Save Answer & Move to Next Question

$\phi = xy$

$$\iiint_R \vec{\nabla}\phi \, dv$$

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

$$\iiint_R \left[\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)(xy)\right] dv$$

$$\iiint_R \left(x\frac{\partial}{\partial x}y\,\hat{i} + y\frac{\partial}{\partial x}x\,\hat{i} + x\frac{\partial}{\partial y}y\,\hat{j} + y\frac{\partial}{\partial y}x\,\hat{j} + x\frac{\partial}{\partial z}y\,\hat{k} + y\frac{\partial}{\partial z}x\,\hat{k}\right) dv$$

$$\iiint_R \left(0 + y\hat{i} + x\hat{j} + 0 + 0 + 0\right) dv$$

$$\iiint_R (y\hat{i} + x\hat{j})\, dv$$

MTH622:Grand Quiz

Question # 2 of 30 (Start time: 10:56:40 AM, 29 December 2020)

The volume element dV in Cylindrical polar coordinates is

Select the correct option

○	$dV = \rho d\rho d\phi$
○	$dV = \rho d\rho d\phi dz$

Type here to search

Question # 25 of 30 (Start time: 02:15:34 PM, 29 December 2020) Total Marks:

For $\varphi = zy$, the integral $\int\int\int_R \vec{\nabla}\varphi dV$, becomes

Select the correct option

Reload Math Equations

- $\int\int\int_R (z\hat{j} + y\hat{k})dV$
- $\int\int\int_R (y\hat{j} + z\hat{k})dV$

Click to Save Answer & Move to Next Question

Question # 1 of 30 (Start time: 11:59:44 AM, 29 December 2020) Total Marks: 1

$$\nabla . V \neq V . \nabla$$

Select the correct option

Reload Math Equations

○	True
○	False

Click to Save Answer & Move to Next Question

Question # 23 of 30 (Start time: 11 17 23 AM, 29 December 2020)

Evaluate $\int \varphi d\vec{r}$ for $\varphi = e^x$

Select the correct option

○	$e^x yz(\hat{i} + \hat{j} + \hat{k})$
○	$e^x \hat{i} + e^x y\hat{j} + e^x z\hat{k}$

Type here to search

Question # 3 of 30 (Start time: 12:01:28 PM, 29 December 2020) Total Mark

The divergence operator when applied to the quantity of a vector field gives a ________ field.

Select the correct option

- Vector
- Scalar

stokes theorem uses

www.khanacademy.org › math › sto...

Stokes' theorem examples (article) | Khan Academy

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How do you know when to use Stokes Theorem?

Which of the following is Stokes Theorem?

Kelvin–Stokes theorem

It is also sometimes known as the **curl theorem**. The classical Kelvin–Stokes theorem relates the surface integral of the curl of a vector field over a surface Σ in Euclidean three-space to **the line integral of** the vector field over its boundary.

en.m.wikipedia.org › wiki › Stokes'...

3:37 p.m. 47%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 1 of 30 (Start time: 03:37:50 PM, 29 December 2020)

Total Marks: 1

The identity $\oint_C \varphi d\vec{r} = -\int\int_S \vec{\nabla}\varphi \times d\vec{S}$ is

Select the correct option

Reload Math Equations

○	true
○	false

Click to Save Answer & Move to Next Question

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 7 of 30 (Start time: 02:28:10 PM, 29 December 2020)

Total Marks: 1

The unit normal to the surface $yz - 1 = 0$ is

Select the correct option

Reload Math Equations

- $\frac{1}{\sqrt{y^2+z^2}}(y\hat{i} + z\hat{k})$
- $\frac{1}{\sqrt{y^2+z^2}}(y\hat{j} + z\hat{k})$
- $\frac{1}{\sqrt{y^2+z^2}}(z\hat{j} + y\hat{k})$
- does not exist

Click to Save Answer & Move to Next Question

4G 63% 2:51 pm

Quiz
quiz.vu.edu.pk

MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 1 of 30 (Start time: 02:51:31 PM, 29 December 2020)

Total Marks: 1

If $\vec{v} = \hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

Reload Math Equations

○	solenoidal
○	non solenoidal

Click to Save Answer & Move to Next Question

Question # 11 of 30 (Start time: 02:34:36 PM, 29 December 2020)

If $\vec{A}$ is a differentiable vector function and φ is differentiable scalar functions of position (x, y, z), then $\left(\vec{A} \times \nabla\right) \varphi =$

Select the correct option

- $\vec{A}.\nabla\varphi$
- $\nabla\left(\vec{A} \times \varphi\right)$
- $\vec{A} \times \varphi$

Question # 27 of 30 (Start time: 02:17:36 PM, 29 December 2020) **Total Marks: 1**

Stokes' theorem is a special case of Green's theorem when applied to a region in the xy-plane.

Select the correct option

Reload Math Equations

○	true
○	false

Click to Save Answer & Move to Next Question

Question # 7 of 30 (Start time: 12:04:56 PM, 29 December 2020)

Gradient of a scalar function represents a ________ vector to the surface.

Select the correct option

- Tangent
- Secant
- Normal
- Unit

MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 17 of 30 (Start time: 03:12:23 PM, 29 December 2020)

Total Marks: 1

The Laplace operator ∇^2 is a ________ operator

Select the correct option

Reload Math Equations

○	Scalar
○	Vector

Click to Save Answer & Move to Next Question

Quiz

https://quiz.vu.edu.pk/QuizQuestio...

MC200205156: NIGHAT NOREEN

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 10 of 30 (Start time: 04:01:32 PM, 29 December 2020)

Total Marks: 1

The _______ of the gradient of φ is zero.

Select the correct option

- Divergence
- Curl
- Gradient
- Scalar field

Click to Save Answer & Move to Next Question

MC200203833: FAIZAN HASSAN

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 13 of 30 (Start time: 03:51:00 PM, 29 December 2020)

Total Marks: 1

__________ is the way of finding derivative in a specific direction, other than coordinate axes.

Select the correct option

- Differentiation
- Directional derivative
- Partial derivative
- nth derivative

Question # 1 of 30 (Start time: 10:55:19 AM, 29 December 2020)

If $\vec{A} = \hat{i}$ then $\int_0^1 \int_0^1 \int_0^1 \vec{A}\, dxdydz =$

Select the correct option

- $-\hat{i}$
- $\hat{i}$
- $3\hat{i}$
- $-3\hat{i}$

3:09

https://quiz.vu.edu.pk/QuizQu

________ is the example of scalar field.

- Velocity
- Displacement
- Acceleration
- Pressure

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quiz.vu.edu.pk/QuizQuestion.aspx?ver=beb8ad5b-4cef-40dc-adc1-2be...

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MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM, 2

Question # 12 of 30 (Start time: 03:37:04 PM, 29 December 2020)

The relation between spherical and cartesian coordinates is

Select the correct option

Reload

○	$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\sin\theta$
○	$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$

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MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 03

Question # 14 of 30 (Start time: 03:39:21 PM, 29 December 2020)

The set of all values of scalar point function φ in R together forms a_______.

Select the correct option

○	scalar field
○	scalar vector
○	scalar function
○	vector function

3:48 p.m. 46%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 11 of 30 (Start time: 03:48:36 PM, 29 December 2020)

Total Marks: 1

The value of $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ for the curve $x = t^2$, $y = 2t - 1$ and $z = 3t^2$ is

Select the correct option

Reload Math Equations

○	$d\vec{r} = (2t + 2 + 0t)dt$
○	$d\vec{r} = (2t\hat{i} + 2\hat{j} + 6t\hat{k})dt$
○	$d\vec{r} = t\hat{i} + 2\hat{j} + 0t\hat{k}$
○	$d\vec{r} = (t^2\hat{i} + 2t\hat{j} + 3t^2\hat{k})dt$

Click to Save Answer & Move to Next Question

Scanned with CamScanner

Question # 7 of 30 (Start time: 11:01:08 AM, 29 December 2020)

Kinetic energy of a body is due to its motion and cannot be transferred into other forms of energies.

Select the correct option

O	true
O	false

Type here to search

MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Time L

Quiz Start Time: 03:26 PM, 29 De

Question # 8 of 30 (Start time: 03:32:01 PM, 29 December 2020)

If a scalar function φ satisfies the Laplace equation

$$\nabla^2\varphi = 0$$

in a Cartesian region R, then φ is said to be a ______ function in the region R.

Select the correct option

Reload Math

○	borel
○	harmonic ✓
○	baire
○	lipschitz

MC200205156: NIGHAT NOREEN

Time Left 79 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 2 of 30 (Start time: 03:55:13 PM, 29 December 2020) Total Marks: 1

If φ and ψ are scalar point functions then $\nabla(\varphi\psi) =$

Select the correct option

Reload Math Equations

○	$\varphi\nabla\psi - \psi\nabla\varphi$
○	$\nabla\varphi \times \nabla\psi$
○	$\nabla\varphi + \nabla\psi$
○	$\varphi\nabla\psi + \psi\nabla\varphi$

Click to Save Answer & Move to Next Question

2:54

Quiz
quiz.vu.edu.pk

MC200204419: ABDUL HAFEEZ

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 27 of 30 (Start time: 02:55:14 PM, 29 December 2020)

Total Marks: 1

$\int_0^1 \int_0^1 \int_0^1 \vec{\nabla} \cdot \hat{i} \, dx dy dz =$

Select the correct option

Reload Math Equations

- ○ -1
- ○ 0
- ○ 1
- ○ 3

Click to Save Answer & Move to Next Question

MC190401710: MALIK MUHAMMAD QASIM

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 18 of 30 (Start time: 02:36:23 PM, 29 December 2020)

Total Marks: 1

Given $\vec{F} = \hat{j} + 4\hat{k}$ and $\vec{r} = 2\hat{k} + \hat{j}$. Find the work done W.

Select the correct option

Reload Math Equations

○	6 units
○	9 units

WhatsApp 4:04 p.m.

1236 messages from 14 chats

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622 Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 25 of 30 (Start time: 04:04:29 PM, 29 December 2020)

Total Marks: 1

If $\vec{T}$ is a unit vector to a space curve C and S is the arc length then $\frac{d\vec{T}}{dS}$ is

Select the correct option

Reload Math Equations

○	parallel to $\vec{T}$
○	perpendicular to $\vec{T}$

3:51 p.m. 45%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 14 of 30 (Start time: 03:51:46 PM, 29 December 2020)

Total Marks: 1

The necessary and sufficient condition that $\oint_C \vec{A}.d\vec{r} = 0$ for every closed curve C is that

Select the correct option

Reload Math Equations

○	$\nabla.\vec{A} = 0$
○	$\nabla \times \vec{A} = 0$

Click to Save Answer & Move to Next Question

MC190403973: LAILA SAIF

MTH622:Grand Quiz

Question # 4 of 30 (Start time: 11:11:41 AM, 29 December 2020)

Evaluate $\int_0^1 \int_0^1 \int_0^1 x \, dx dy dz =$

Select the correct option

- $\frac{1}{2}$
- $\frac{-1}{2}$

WhatsApp

1160 messages from 14 chats

3:39 p.m.

MC200203833: FAIZAN HASSAN

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 3 of 30 (Start time: 03:39:58 PM, 29 December 2020)

Total Marks: 1

Evaluate $\int_c (x^2 dx + y^2 dy)$ along the line segment joining $(1,0)$ to $(1,1)$.

Select the correct option

Reload Math Equations

○	$\frac{1}{3}$
○	$\frac{1}{2}$

Question # 6 of 30 (Start time: 11:00:17 AM, 29 December 2020)

The Stoke's theorem can be used to find which of the following?

Select the correct option

○	Surface area enclosed by a function in the given region
○	Volume enclosed by a function in the given region
○	Linear distance
○	Curl of the function

Type here to search

MC190403973: LAILA SAIF

MTH622:Grand Quiz

Quiz Start Time: 11

Question # 5 of 30 (Start time: 11:13:19 AM, 29 December 2020)

$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ is called as the differential ______ vector.

Select the correct option

- Displacement
- Normal
- Unit
- Velocity

MC190401710: MALIK MUHAMMAD QASIM

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 20 of 30 (Start time: 02:38:31 PM, 29 December 2020)

Total Marks: 1

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = y\hat{j}$ becomes

Select the correct option

Reload Math Equations

○	$\oint_C (y\,dz\,\hat{i} + y\,dy\,\hat{k})$
○	$\oint_C (-y\,dz\,\hat{i} + y\,dx\,\hat{k})$

Click to Save Answer & Move to Next Question

MC190401710: MALIK MUHAMMAD QASIM

Time Left 83 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 202

Question # 23 of 30 (Start time: 02:41:42 PM, 29 December 2020)

Total Marks:

The curl of a curl of a vector gives a _______.

Select the correct option

- scalar
- vector
- zero vector
- non of these

- Pressure
- Temperature
- Velocity
- Distance

MTH622:Grand Quiz

Question # 16 of 30 (Start time: 11:09:57 AM, 29 December 2020)

If $\vec{V}$ is a solenoid vector point function then

Select the correct option

○	
○	
○	
○	

Type here to search

Time Left

MC190406995: AKHTAR HASSAN NOORI

Quiz Start Time: 03:26 PM, 29 Decem

MTH622:Grand Quiz

Total

Question # 29 of 30 (Start time: 03:55:59 PM, 29 December 2020)

The radial component of velocity for a particle moving in circular path is

Select the correct option

○	constant
○	radius itself
○	variable
○	zero

MC200205156: NIGHAT NOREEN

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 12 of 30 (Start time: 04:03:17 PM, 29 December 2020)

Total Marks: 1

There exists a continuous vector field $\vec{V}$ for a scalar potential function φ that satisfies the relation

Select the correct option

Reload Math Equations

○	$\nabla \times \vec{V} = \varphi$
○	$\vec{V} = -\nabla\varphi$
○	$\vec{V} = \nabla \times \varphi$
○	$\vec{V} = \nabla\varphi$

Click to Save Answer & Move to Next Question

Question # 6 of 30 (Start time: 12:05:19 PM, 29 December 2020)

Total Marks

The Laplace operator ∇^2 is a ________ operator.

Select the correct option

Reload Math Equation

- Scalar
- Vector

Click to Save Answer & Move to Next Question

3:17 pm 55%

Quiz
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MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 21 of 30 (Start time: 03:17:04 PM, 29 December 2020)

Total Marks:

The Laplace operator is ______order differential operator given by the divergence of the gradient of a given function defined over a space R.

Select the correct option

- first
- second
- third
- All of above

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WhatsApp
1164 messages from 14 chats
3:42 p.m.

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 5 of 30 (Start time: 03:42:50 PM, 29 December 2020)

Total Marks: 1

An ellipse is a simple closed curve.

Select the correct option

○	true
○	false

MC200205156: NIGHAT NOREEN

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 11 of 30 (Start time: 04:02:27 PM, 29 December 2020)

Total Marks: 1

If a force acts on a particle of constant mass m and $\vec{v}_1$, $\vec{v}_2$ are velocities at times t_1 and t_2, respectively. Then the work done by the force is obtained as

Select the correct option

Reload Math Equations

- The change in kinetic energies, i.e., $\frac{1}{2}m\vec{v}_2^2 - \frac{1}{2}m\vec{v}_1^2$
- The sum of kinetic energies, i.e., $\frac{1}{2}m\vec{v}_2^2 + \frac{1}{2}m\vec{v}_1^2$

Click to Save Answer & Move to Next Question

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MC190406995: AKHTAR HASSAN NOORI

Time

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM, 29 I

Question # 27 of 30 (Start time: 03:54:30 PM, 29 December 2020)

The directional derivative $\frac{\partial\varphi}{\partial r} = |\nabla\varphi||\hat{a}|\cos\theta$ is zero, when θ = ________.

Select the correct option

Reload Ma

○	0°
○	180°
○	90° ✓
○	-180°

Question # 23 of 30 (Start time: 12:46:32 PM, 29 December 2020) Total Marks: 1

A field which is not irrotational is sometimes called a ______field.

Select the correct option

- Vortex
- Non rotational
- Solenoid
- Tangential

Click to Save Answer & Move to Next Question

3:53 p.m. 45%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 15 of 30 (Start time: 03:53:06 PM, 29 December 2020)

Total Marks: 1

Both double and triple integrals can be used to calculate

Select the correct option

Reload Math Equations

○	area under the curve
○	length of curve
○	volume of the given object
○	None of these

8:33

quiz.vu.edu.pk/QuizQu

MC200201687: MUHAMMAD NAUMAN

MTH622 Grand Quiz

Quiz Start Time: 08:32 AM

Question # 2 of 30 (Start time: 08:33:08 AM, 29 December 2020)

The identity $\oint_C \vec{B} \cdot d\vec{r} = \iint_S (\hat{n} \times \vec{\nabla}) \times \vec{B} dS$ is

Select the correct option

○	true
○	false

Jazz 21.2K/s 47% 2:28 PM

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MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 8 of 30 (Start time: 02:28:48 PM, 29 December 2020)

Total Marks: 1

The gradient of $x\hat{i} + y\hat{j} + z\hat{k}$ is ______

Select the correct option

Reload Math Equations

O	0
O	1
O	2
O	3

Click to Save Answer & Move to Next Question

12:58 4G 18

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MC200204967: KASHIF MEHMOOD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 12:56 PM, 29 December 2020

Question # 3 of 30 (Start time: 12:58:26 PM, 29 December 2020)

Total Marks:

The necessary and sufficient condition that $\int_C \vec{A}.\,d\vec{r} = 0$ for every closed curve C is that

Select the correct option

Reload Math Equations

- $\nabla.\vec{A} = 0$
- $\nabla \times \vec{A} = 0$

Click to Save Answer & Move to Next Question

Jazz 0K/s 35% 2:46 PM

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MC190401710: MALIK MUHAMMAD QASIM

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 29 of 30 (Start time: 02:45:54 PM, 29 December 2020)

Total Marks: 1

$\nabla \times \vec{A}$ is used to denote

Select the correct option

Reload Math Equations

○	a. Curl $\vec{A}$
○	b. Rot $\vec{A}$
○	c. Gradient $\vec{A}$
○	d. Both a and b

Click to Save Answer & Move to Next Question

Quiz

MC200203833: FAIZAN HASSAN

Time Left 83 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 4 of 30 (Start time: 03:41:25 PM, 29 December 2020) Total Marks: 1

Inertia is directly related with mass of the body.

Select the correct option

- true
- false

MC190400548: ANSA SAID

MTH622 Grand Quiz

Question # 21 of 30 (Start time: 11:15:25 AM, 29 December 2020)

Let $\vec{A}$ be a given vector point function which is defined and continuous in a closed region R. Then the volume integral of $\vec{A}$ over the region R can be defin

Select the correct option

O	$\int\int\int_R \vec{A}.\, dV$
O	$\int\int\int_R \vec{A} dV$
O	$\int\int\int_R \vec{A} \times dV$
O	$\int\int\int_R \nabla \vec{A} dV$

Type here to search

2:52

Quiz
quiz.vu.edu.pk

MC200204419: ABDUL HAFEEZ

Time Left 85 sec(s)

MTH622 Grand Quiz

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 24 of 30 (Start time: 02:52:50 PM, 29 December 2020)

Total Marks: 1

The mathematical perception of the gradient is said to be

Select the correct option

- Tangent
- Chord
- Slope
- Arc

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MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Tim

Question # 24 of 30 (Start time: 03:51:07 PM, 29 December 2020)

The Laplace operator ∇^2 is a __________ operator.

Select the correct option

○	Scalar
○	Vector

MTH622:Grand Quiz

Question # 9 of 30 (Start time: 12:07:50 PM, 29 December 2020)

The unit normal to the surface $x + z - 1 = 0$ is

Select the correct option

- $\frac{1}{\sqrt{2}}(\hat{k} + \hat{j})$
- $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$
- $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$
- does not exist

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MTH622:Grand Quiz

Question # 8 of 30 (Start time: 12:06:23 PM, 29 December 2020)

The value of $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ for the curve $x = t + 1$, $y = 2t^2$ and $z = t^3$ is

Select the correct option

- $d\vec{r} = (1 + 4t + 3t^2)dt$
- $d\vec{r} = (\hat{i} + t\hat{j} + t^2\hat{k})dt$
- $d\vec{r} = t\hat{i} + 4t\hat{j} + 3t^2\hat{k}$
- $d\vec{r} = (\hat{i} + 4t\hat{j} + 3t^2\hat{k})dt$

...ized Quiz **Total Questions : ...**

...lease read the following instructions carefully!

1. Quiz will be based upon **Multiple Choice Questions** (MCQs).
2. You have to attempt the quiz online. You can start attempting the quiz any time within given date(s) of a particular subject by clicking the link for Quiz ... VULMS.
3. Each question has a fixed time of 90 seconds. So you have to save your answer before 90 seconds. But due to unstable internet speeds, it is recommend... that you should save your answer within 60 seconds. While attempting a question, keep an eye on the remaining time.
4. Attempting quiz is unidirectional. Once you move forward to the next question, you can not go back to the previous one. Therefore before moving to the ne... question, make sure that you have selected the best option.
5. **DO NOT** press **Back Button / Backspace Button** while attempting a question, otherwise you will lose that question.
6. **DO NOT** refresh the page unnecessarily, **specially when following messages appear**
   - Saving...
   - Question Timeout: Now loading next question.
7. **Javascript MUST be enabled** in your browser; otherwise you will not be able to attempt the quiz.
8. If for any reason, you lose access to internet (like power failure or disconnection of internet), you will be able to attempt the quiz again from the question ne... to the last shown question. But remember that you have to complete the quiz before expiry of the deadline.
9. If any student failed to attempt the quiz in given time then no re-take or offline quiz will be held.

Start Quiz

...d2c25d-c6e4-4b27-a540-f65ed2c1acc0

Apps Gmail YouTube Maps

MC190403548 ANSA SAID

MTH622 Grand Quiz

Question # 8 of 30 (Start time: 11:01:42 AM, 29 December 2020)

If $\int_{P_1}^{P_2} \vec{V}.d\vec{r}$ is independent of the path joining any two points P_1 and P_2 in a given region is that $\oint_C \vec{V}.d\vec{r} = 0$ for all _____ path C in the region

Select the correct option

○	open
○	closed
○	circular
○	Both open and close

Type here to search

Jazz 15.6K/s 4G 51% 2:24 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 2 of 30 (Start time: 02:24:00 PM, 29 December 2020)

Total Marks: 1

$\hat{j} \times \vec{\nabla}\varphi$ for $\varphi = x$ becomes

Select the correct option

Reload Math Equations

○	$\hat{k}$
○	$-\hat{k}$

Click to Save Answer & Move to Next Question

MC190403548: ANSA SAID

MTH622:Grand Quiz

Question # 25 of 30 (Start time: 11:18:29 AM, 29 December 2020)

If $\vec{A}$ is a differentiable vector function and φ is differentiable scalar functions of position (x, y, z), then $\left(\vec{A} \times \nabla\right) \varphi =$

Select the correct option

O	$\vec{A}.\nabla\varphi$
O	$\nabla\left(\vec{A} \times \varphi\right)$
O	$\vec{A} \times \varphi$
O	$\vec{A} \times (\nabla\varphi)$

Type here to search

Question # 28 of 30 (Start time: 02:18:27 PM, 29 December 2020) Total Marks: 1

Evaluate $\int_0^1 \int_0^1 \int_0^1 y \, dx dy dz =$

Select the correct option

Reload Math Equations

○	$\frac{1}{2}$
○	$\frac{-1}{2}$

Click to Save Answer & Move to Next Question

Jazz 2K/s 49% 2:26 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 5 of 30 (Start time: 02:25:58 PM, 29 December 2020)

Total Marks: 1

If $\vec{v} = x\hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

Reload Math Equations

○	solenoidal
○	non solenoidal

Click to Save Answer & Move to Next Question

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MC190406995: AKHTAR HASSAN NOORI

Time Left 87 sec(

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM, 29 Decembe

Question # 16 of 30 (Start time: 03:41:26 PM, 29 December 2020)

Total M

The two dimensional Laplace operators in polar coordinate (r, θ) can be expressed as $\nabla^2\varphi =$

Select the correct option

Reload Math Equat

○	$\frac{\partial \varphi}{\partial \theta} + \frac{1}{r}\frac{\partial \varphi}{\partial \theta} + \frac{1}{r^2}\frac{\partial^2 \varphi}{\partial \theta^2}$
○	$\frac{\partial^2 \varphi}{\partial r^2} + \frac{1}{r}\frac{\partial \varphi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \varphi}{\partial \theta^2}$
○	$\frac{1}{r}\frac{\partial \varphi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \varphi}{\partial \theta^2}$
○	$\frac{\partial^2 \varphi}{\partial \theta^2} + \frac{1}{r^2}\frac{\partial^2 \varphi}{\partial \theta^2}$

MC190403973: LAILA SAIF

MTH622:Grand Quiz

Question # 3 of 30 (Start time: 11:09:48 AM, 29 December 2020)

The set of all values of scalar point function φ in R together forms a ______.

Select the correct option

- scalar field
- scalar vector
- scalar function
- vector function

quiz.vu.edu.pk/QuizQuestion.aspx?ver=aa28c554-2981-4fd1-af6a-5a30...

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MC190406995 AKHTAR HASSAN NOORI

MTH622 Grand Quiz

Quiz Start Time 03 26 PM, 2

Question # 6 of 30 (Start time 03 30 06 PM, 29 December 2020)

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = \hat{k}$ becomes

Select the correct option

Reload

○	$\oint_C (dy\hat{i} - dx\hat{j})$
○	$\oint_C (dz\hat{i} - dx\hat{k})$

quiz.vu.edu.pk/QuizQuestion.aspx?ver=533e22b8-49f5-4c7f-

Apps Gmail YouTube Maps Translate New Tab

MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time

Question # 19 of 30 (Start time: 03:44:47 PM, 29 December 2020)

For $\varphi = xy$, the integral $\int\int\int_R \vec{\nabla}\varphi dV$

Select the correct option

○	$\int\int\int_R (x\hat{i} + y\hat{j})dV$
○	$\int\int\int_R (y\hat{i} + x\hat{j})dV$

Question # 30 of 30 (Start time: 12:52:53 PM, 29 December 2020) Total Marks: 1

If $\vec{F}$ is a conservative vector field then $\int_C \vec{F} . d\vec{r}$ is ______of path.

Select the correct option

Reload Math Equations

O	dependent
O	independent

Click to Save Answer & Move to Next Question

3:29 pm 52%

Quiz
quiz.vu.edu.pk

MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 28 of 30 (Start time: 03:29:08 PM, 29 December 2020)

Total Marks: 1

The surface integral of a vector field is called as the _______ of vector field through the surface.

Select the correct option

- Gradient
- Flux
- Divergence
- Curl

Click to Save Answer & Move to Next Question

MC200204419: ABDUL HAFEEZ

MTH622:Grand Quiz

Time Left 87 sec(s)

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 23 of 30 (Start time: 02:51:50 PM, 29 December 2020)

Total Marks:

Gradient of a scalar function represents a ________ vector to the surface.

Select the correct option

○	Tangent
○	Secant
○	Normal
○	Unit

MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 202

Question # 24 of 30 (Start time: 03:21:20 PM, 29 December 2020)

Total Marks:

A particle moves in a plane in such a way that at any time t its distance from the origin is $r = 2t + 3t^2$ and its angle with positive x-axis is $\theta = \sqrt{t}$. Find the radial and transverse components of velocity

Select the correct option

Reload Math Equations

- $2 + 6t$ and $(\sqrt{t} + \frac{3}{2}t^{\frac{3}{2}})$
- $2 + 6t$ and $\frac{1}{2\sqrt{t}}$

Click to Save Answer & Move to Next Question

Jazz 14K/s 38% 2:42 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 24 of 30 (Start time: 02:42:34 PM, 29 December 2020)

Total Marks: 1

The directional derivative $\frac{\partial\varphi}{\partial s} = |\nabla\varphi||\vec{a}|\cos\theta$ is zero, when θ = ______

Select the correct option

Reload Math Equations

O	0°
O	180°
O	90°
O	-180°

Click to Save Answer & Move to Next Question

2:50

Quiz
quiz.vu.edu.pk

MC200204419: ABDUL HAFEEZ

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 22 of 30 (Start time: 02:50:44 PM, 29 December 2020)

Total Marks: 1

The tangential form of Green's theorem in plane is also called Stokes theorem in the plane

Select the correct option

Reload Math Equations

O	true
O	false

Click to Save Answer & Move to Next Question

If k is curvature of a space curve then radius of curvature is

Select the correct option

Reload Math Equations

○	$\sqrt{k}$
○	$\frac{1}{k}$

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MC190406995: AKHTAR HASSAN NOORI

Time Le

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM, 29 Dec

Question # 20 of 30 (Start time: 03:45:59 PM, 29 December 2020)

If $\nabla . \vec{V} = 0$ everywhere in some region of R, then $\vec{V}$ is called ________point function in the region

Select the correct option

Reload Math

○	zero vector
○	unit vector
○	co-initial vector
○	solenoid vector

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MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 22 of 30 (Start time: 04:02:11 PM, 29 December 2020)

Total Marks: 1

The _______ of the gradient of is zero.

Select the correct option

- Divergence
- Curl
- Gradient
- Scalar field

Jazz 16.6K/s 46% 2:30 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 10 of 30 (Start time: 02:30:12 PM, 29 December 2020)

Total Marks: 1

Green's theorem implies that if $\oint_c M dx + N dy = 0$, then $\frac{\partial N}{\partial y} = \frac{\partial M}{\partial x}$.

Select the correct option

Reload Math Equations

○	true
○	false

Click to Save Answer & Move to Next Question

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https://quiz.vu.edu.pk/QuizQuestion.aspx?ver=f8e7570e-8f4a-4ff6-b2ad-4f7b7c540dee

MC190403973 LAILA SAIF

MTH622:Grand Quiz

Quiz Start Time 11:0...

Question # 28 of 30 (Start time: 11:47:12 AM, 29 December 2020)

If a particle of mass 2 units has position vector $\vec{r} = t^3\hat{i} + 8t\hat{j} - 9t\hat{k}$ then find the force acting on the particle

Select the correct option

○	$12t$
○	$12t\hat{i}$

MTH622:Grand Quiz

Question # 2 of 30 (Start time: 11:59:21 AM, 29 December 2020)

The property $\nabla . \vec{V} = \vec{V} . \nabla$ always holds.

Select the correct option

○	True
○	False

Jazz 16.3K/s 4G 35% 2:46 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 30 of 30 (Start time: 02:46:40 PM, 29 December 2020)

Total Marks: 1

If $\varphi = b$ then $\int_0^a \int_0^a \int_0^a \varphi \, dx dy dz =$

Select the correct option

Reload Math Equations

○	ba
○	ba^3
○	$b\frac{a^2}{3}$
○	$3ab$

Click to Save Answer & Move to Next Question

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 8 of 30 (Start time: 03:45:42 PM, 29 December 2020)

Total Marks: 1

The Green's theorem in plane can be written in tangential form as

Select the correct option

Reload Math Equations

○	$\oint_C \vec{B}.\hat{n}dS = \int\int_S \vec{\nabla}.\vec{A}dR$
○	$\oint_C \vec{A}.d\vec{r} = \int\int_R(\vec{\nabla}\times\vec{A}).\hat{k}dR$

Click to Save Answer & Move to Next Question

MTH622:Grand Quiz

Question # 1 of 30 (Start time: 03:07:19 PM, 29 December 2020)

The unit normal to the surface $x = 0$ is

Select the correct option

○	$\pm\hat{j}$
○	$\pm\hat{i}$
○	$\pm\hat{k}$
○	does not exist

MC190401710: MALIK MUHAMMAD QASIM

Time Left 69 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 21 of 30 (Start time: 02:39:40 PM, 29 December 2020)

Total Marks: 1

________ is the example of scalar field.

Select the correct option

- Velocity
- Displacement
- Acceleration
- Pressure

Click to Save Answer & Move to Next Question

Module No. 71

Green's Theorem in th
of Stokes'

Green's theorem can be expressed in the plan
tangential form or normal forms of Green's theor
The tangential form of Green's theorem is also
This generalize form of Green's theorem in plan
we can say that Green' theorem is a special case
the xy-plane.

Question # 20 of 30 (Start time: 11:14:34 AM, 29 December 2020)

Unit circle is not a simple closed curve.

Select the correct option

○	true
○	false

Type here to search

4G 60% 3:00 pm

Quiz
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MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 8 of 30 (Start time: 03:00:58 PM, 29 December 2020)

Total Marks:

$\hat{i} \times \vec{\nabla}\varphi$ for $\varphi = x$ becomes

Select the correct option

Reload Math Equations

O	$\vec{0}$
O	$\hat{j}$

Click to Save Answer & Move to Next Question

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MC190406995: AKHTAR HASSAN NOORI

Time Left 84 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM, 29 December 202

Question # 25 of 30 (Start time: 03:51:58 PM, 29 December 2020)

Total Marks:

If $F = xyi - yj$. Then the integral for work done by the force on the curve C in the xy plane, $y = 2x$, from (0, 0) to (1, 2) is

Select the correct option

Reload Math Equations

- $\int_0^1 (2x^2dx - 4xdx)$
- $\int_1^2 (2xdx + 4xdx)$
- $\int_0^1 (xdx - ydx)$
- $\int_0^2 (2x^2dx + 4xdx)$

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MC190203990: MUHAMMAD REHAN MURAD

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 202

Question # 3 of 30 (Start time: 02:54:26 PM, 29 December 2020)

Total Marks:

An integral where the function is evaluated along a _______ is called line integral.

Select the correct option

- line
- curve
- scalar point
- vector field

Click to Save Answer & Move to Next Question

Virtual University of × | Quiz List × | Quiz × +

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MC190406995: AKHTAR HASSAN NOORI

Time

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM, 29 D

Question # 18 of 30 (Start time: 03 44 06 PM, 29 December 2020)

All laws of mechanics are dependent on frame of reference.

Select the correct option

○	true
○	false

Question # 11 of 30 (Start time: 11:04:59 AM, 29 December 2020)

An example of path independence is work done by the gravitation force.

Select the correct option

○	True
○	False

Type here to search

MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 03

Question # 10 of 30 (Start time: 03:34:52 PM, 29 December 2020)

A curl has ________ divergence.

Select the correct option

○	zero
○	non zero

Question # 13 of 30 (Start time: 11:06:47 AM, 29 December 2020)

If $\vec{v} = \hat{i}$, then the vector field $\vec{v}$ is

Select the correct option

○	solenoidal
○	non solenoidal ✓

Type here to search

3:15 pm 56%

Quiz
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MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 19 of 30 (Start time: 03:15:13 PM, 29 December 2020)

Total Marks:

$\int_C \vec{A}.d\vec{r}$ for $\vec{A} = 2\hat{i}$ becomes

Select the correct option

Reload Math Equations

○	$2\int_C \hat{i}dx$
○	$2\int_C dx$

Click to Save Answer & Move to Next Question

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MC190406995: AKHTAR HASSAN NOORI

Tin

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM, 2

Question # 26 of 30 (Start time: 03:53:14 PM, 29 December 2020)

If $\vec{A}$ is irrotational and $\nabla \times \vec{A} = (c-1)\hat{i} + (a-4)\hat{j} + (b-2)\hat{k}$. Then the value of a must be

Select the correct option

Upload M

○	2
○	0
○	1 ✓
○	4

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Jazz 155B/s 50% 2:25 PM

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MC190401710: MALIK MUHAMMAD QASIM

Time Left 81 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 4 of 30 (Start time: 02:25:13 PM, 29 December 2020)

Total Marks:

Classical mechanics deals with the

Select the correct option

- Statics objects only
- microscopic objects only
- macroscopic objects only
- None of these

16:00 66%

Quiz
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MC200205156: NIGHAT NOREEN

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 9 of 30 (Start time: 04:00:54 PM, 29 December 2020)

Total Marks: 1

$$\nabla\varphi . \hat{j} = \_\_\_\_\_\_.$$

Select the correct option

Reload Math Equations

○	$\frac{\partial\varphi}{\partial x}$
○	$\frac{\partial\varphi}{\partial y}$
○	$\frac{\partial\varphi}{\partial z}$
○	$\frac{\partial\varphi}{\partial y}\hat{j}$

Click to Save Answer & Move to Next Question

Quiz
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MC190203990: MUHAMMAD REHAN MURAD

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 6 of 30 (Start time: 02:57:56 PM, 29 December 2020)

Total Marks: 1

Evaluate $\int_0^1 \int_0^1 \int_0^1 x \, dx dy dz =$

Select the correct option

Reload Math Equations

- $\frac{1}{3}$
- $\frac{1}{2}$

Click to Save Answer & Move to Next Question

MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 7 of 30 (Start time: 02:58:39 PM, 29 December 2020)

Total Marks: 1

A particle moves in a plane in such a way that at any time t its distance from the origin is $r = 3t^2$ and its angle with positive x-axis is $\theta = \sqrt{t}$. Find the radial component of acceleration

Select the correct option

Reload Math Equations

○	$6t - r\frac{1}{4t}$
○	$6 - r\frac{1}{4t}$

Click to Save Answer & Move to Next Question

MTH622 Grand Quiz

Question # 12 of 30 (Start time: 11:05:38 AM, 29 December 2020)

If $\vec{A} = \hat{i}$ then $\int_1^0 \int_1^0 \int_1^0 \vec{A}\, dxdydz =$

Select the correct option

- $-3\hat{i}$
- $\hat{i}$
- $3\hat{i}$
- $-\hat{i}$

Type here to search

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2:56

Quiz
quiz.vu.edu.pk

MC200204419: ABDUL HAFEEZ

Time Left 83 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 29 of 30 (Start time: 02:56:57 PM, 29 December 2020)

Total Marks:

$\vec{j} \times \vec{\nabla}\phi$ for $\phi = j$ becomes

Select the correct option

Reload Math Equations

O	$\vec{0}$
O	$\vec{i}$

Question # 6 of 30 (Start time: 12:04:14 PM, 29 December 2020)

The second vector form of Green's theorem is also called

Select the correct option

Normal form of Green's theorem

tangential form of Green's theorem

MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 2 of 30 (Start time: 02:52:09 PM, 29 December 2020)

Total Marks: 1

$\hat{j} \times \bar{\nabla}\varphi$ for $\varphi = j$ becomes

Select the correct option

Reload Math Equations

○	$\vec{0}$
○	$\hat{i}$

Click to Save Answer & Move to Next Question

MTH622:Grand Quiz

Question # 9 of 30 (Start time: 12:07:50 PM, 29 December 2020)

The unit normal to the surface $x + z - 1 = 0$ is

Select the correct option

○	$\frac{1}{\sqrt{2}}(\hat{k} + \hat{j})$
○	$\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$
○	$\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$
○	does not exist

Jazz 683B/s 43% 2:33 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 15 of 30 (Start time: 02:33:48 PM, 29 December 2020)

Total Marks: 1

Greens theorem states that if R is simply connected region of the xy-plane bounded by a closed curve C and if M and N are continuous functions of x and y having continuous derivatives in R then

Select the correct option

Reload Math Equations

○	$\oint_C Mdy + Ndx = \int\int_R (\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y})dxdy$
○	$\oint_C Mdx + Ndy = \int\int_R (\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y})dxdy$

Click to Save Answer & Move to Next Question

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Question # 14 of 30 (Start time: 11:07:50 AM, 29 December 2020)

The generalization of Green's theorem as Gauss's divergence theorem is also called the normal form of Green's theorem

Select the correct option

○	True
○	false

Type here to search

MC190203990: MUHAMMAD REHAN MURAD

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 11 of 30 (Start time: 03:06:10 PM, 29 December 2020)

Total Marks: 1

The radial and transverse components of velocity are

Select the correct option

Reload Math Equations

○	$\dot{r},\ r\dot{\theta}$
○	$\dot{r},\ \theta\dot{\theta}$

Click to Save Answer & Move to Next Question

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MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz **Quiz Start Time: 03:26 PM.**

Question # 11 of 30 (Start time: 03:36:05 PM, 29 December 2020)

If $\nabla\varphi = -\hat{i} + 2\hat{j} - \hat{k}$, then $|\nabla\varphi|$ = ________.

Select the correct option

Reload

○	2
○	6
○	3
○	$\sqrt{6}$

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MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 19 of 30 (Start time: 02:37:44 PM, 29 December 2020)

Total Marks: 1

If $\varphi = 1$ then $\int_1^0 \int_1^0 \int_1^0 \varphi \, dx dy dz =$

Select the correct option

Reload Math Equations

O	1
O	-1
O	3
O	-3

Click to Save Answer & Move to Next Question

57% 3:13 pm

Quiz
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MC190203990: MUHAMMAD REHAN MURAD

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 18 of 30 (Start time: 03:12:59 PM, 29 December 2020)

Total Marks: 1

If $F = x\vec{i} + y\vec{j}$. Then the integral for work done by the force on the curve C in the xy plane, $y = x$, from (0, 0) to (2, 2) is

Select the correct option

Reload Math Equations

○	$\int_0^2 (xdx - 2xdx)$
○	$\int_2^2 (xdx)$
○	$\int_0^2 (ydx - xdx)$
○	$\int_0^2 (2xdx)$

Click to Save Answer & Move to Next Question

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MC190403973 LAILA SAIF

MTH622:Grand Quiz

Time Left 83

Question # 26 of 30 (Start time: 11:44:53 AM, 29 December 2020)

Quiz Start Time: 11:07 AM, 29 December 2020

If $F = x\hat{i} + y\hat{j}$. Then the integral for work done by the force on the curve C in the xy plane $y = x$, from (0, 0) to (2, 2) is

Total Marks: 1

Select the correct option

Reload Math Equations

- ○ $\int_0^2 (x\,dx - 2x\,dx)$
- ○ $\int_2^2 (x\,dx)$
- ○ $\int_0^2 (y\,dx - x\,dx)$
- ○ $\int_0^2 (2x\,dx)$

Google Chrome

MC190401710: MALIK MUHAMMAD QASIM

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 3 of 30 (Start time: 02:24:39 PM, 29 December 2020)

Total Marks: 1

Evaluate $\int_c (x+y)dy$ along the line segment joining $(0,0)$ to $(0,1)$

Select the correct option

Reload Math Equations

- $\frac{1}{3}$
- $\frac{1}{2}$

Click to Save Answer & Move to Next Question

BC180404025: SIDRA JALAL

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:14 PM, 29 December 2020

Question # 8 of 30 (Start time: 02:22:14 PM, 29 December 2020)

Total Marks: 1

If $\vec{F}$ is the force on a particle move along C, then line integral $\int_C \vec{F}.d\vec{r}$ show the ______ by the force.

Select the correct option

Reload Math Equations

○	Friction
○	Work done
○	Momentum
○	Torsion

Click to Save Answer & Move to Next Question

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 16 of 30 (Start time: 03:54:20 PM, 29 December 2020)

Total Marks: 1

The divergence operator when applied to the quantity of a vector field gives a __________ field.

Select the correct option

O	Vector
O	Scalar

MC200204419: ABDUL HAFEEZ

Time Left 84 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 30 of 30 (Start time: 02:58:09 PM, 29 December 2020)

Total Marks: 1

If $\vec{A} = \hat{i}$ then $\int_1^0 \int_1^0 \int_1^0 \vec{A}\, dx\, dy\, dz =$

Select the correct option

Reload Math Equations

- $-3\hat{i}$
- $\hat{i}$
- $3\hat{i}$
- $-\hat{i}$

50% 3:31 pm

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MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 30 of 30 (Start time: 03:31:41 PM, 29 December 2020)

Total Marks: 1

Gauss divergence theorem is used to derive equation of which phenomena

Select the correct option

Reload Math Equations

- flow of fluids
- heat conduction
- wave propagation
- all of these

Click to Save Answer & Move to Next Question

MC200204419: ABDUL HAFEEZ

Time Left 85 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 21 of 30 (Start time: 02:49:15 PM, 29 December 2020)

Total Marks: 1

If $\varphi = 1$ then $\int_0^1 \int_0^1 \int_0^1 \varphi \, dx dy dz =$

Select the correct option

Reload Math Equations

O	1
O	0
O	3
O	-3

Click to Save Answer & Move to Next Question

12:57

quiz.vu.edu.pk/QuizQu

MC200204967: KASHIF MEHMOOD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 12:56 PM, 29 December 2020

Question # 2 of 30 (Start time: 12:57:19 PM, 29 December 2020)

Total Marks: 1

An example of path independence is work done by the gravitation force.

Select the correct option

- True
- False

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MTH622 Grand Quiz Quiz Start Time: 03:26

Question # 7 of 30 (Start time: 03:31:00 PM 29 December 2020)

The unit normal to the surface $yz - 1 = 0$ is

Select the correct option

Rel

○	$\frac{1}{\sqrt{y^2+z^2}}(y\hat{i} + z\hat{k})$
○	$\frac{1}{\sqrt{y^2+z^2}}(y\hat{j} + z\hat{k})$
○	$\frac{1}{\sqrt{y^2+z^2}}(z\hat{j} + y\hat{k})$
○	does not exist

15:57 68%

Quiz
https://quiz.vu.edu.pk/QuizQuestio...

MC200205156: NIGHAT NOREEN

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 4 of 30 (Start time: 03:56:53 PM, 29 December 2020)

Total Marks: 1

The directional derivative $\frac{\partial\varphi}{\partial u} = |\nabla\varphi||\hat{u}|\cos\theta$ is zero, when θ = ________

Select the correct option

Reload Math Equations

○	0°
○	180°
○	90°
○	-180°

Click to Save Answer & Move to Next Question

MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 03:26

Question # 30 of 30 (Start time: 03:57:30 PM, 29 December 2020)

If $\vec{v} = b\hat{k}$, then the vector field $\vec{v}$ is

Select the correct option

○	solenoidal
○	non solenoidal

The mathematical perception of the gradient is said to be

Select the correct option

- Tangent
- Chord
- Slope
- Arc

MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 15 of 30 (Start time: 03:10:55 PM, 29 December 2020)

Total Marks:

__________ is the way of finding derivative in a specific direction, other than coordinate axes.

Select the correct option

- Differentiation
- Directional derivative
- Partial derivative
- nth derivative

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MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 0

Question # 28 of 30 (Start time: 03:55:10 PM, 29 December 2020)

If $\nabla\varphi = 4\hat{j} - 3\hat{k}$, then $|\nabla\varphi|$= ________.

Select the correct option

○	2
○	5
○	-5
○	$\sqrt{7}$

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MC200200527: AMINA MANZOOR

TH622:Grand Quiz

Time Left

Quiz Start Time: 11:59 AM, 29 December 2020

uestion # 4 of 30 (Start time: 12:03:46 PM, 29 December 2020)

Total Marks: 1

An ellipse is a simple closed curve.

elect the correct option

- true
- false

stion # 24 of 30 (Start time: 11:18:00 AM, 29 December 2020)

e directional derivative $\frac{\partial \varphi}{\partial s} = |\nabla \varphi||\hat{a}| \cos \theta$ is maximum, when θ = ________

ct the correct option

- 0^o
- 180^o
- 90^o
- 270^o

Type here to search

MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 24 of 30 (Start time: 03:21:20 PM, 29 December 2020)

Total Marks:

A particle moves in a plane in such a way that at any time t its distance from the origin is $r = 2t + 3t^2$ and its angle with positive x-axis is $\theta = \sqrt{t}$. Find the radial and transverse components of velocity

Select the correct option

Reload Math Equations

- $2 + 6t$ and $\left(\sqrt{t} + \frac{3}{2}t^{\frac{3}{2}}\right)$
- $2 + 6t$ and $\frac{1}{2\sqrt{t}}$

Click to Save Answer & Move to Next Question

MTH622:Grand Quiz

Question # 19 of 30 (Start time: 11:13:14 AM, 29 December 2020)

If a scalar function φ satisfies the Laplace equation

in a Cartesian region R, then φ is said to be a ______ function in the region R.

Select the correct option

- borel
- harmonic
- baire
- lipschitz

ement

alled irrotational if $curl v = 0$.

nstants a, b, c so that

$$\vec{A} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)$$

tional.

hat $\vec{A}$ can be expressed as the gradient of a scalar function.

vecto field is

$$\vec{A} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{\imath}$$

or a vector to be irrotational, we have expression

$$\text{curl}\vec{A} = \nabla \times \vec{A} = 0$$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + 2y + az & bx - 3y - z & 4x + cy + 2z \end{vmatrix}$$

$$+ cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z)\Big)\hat{\imath} - \left(\frac{\partial}{\partial x}(4x + cy + 2z) - \frac{\partial}{\partial z}(x + 2y$$

ine integral is used to calculate

Select the correct option

○	Force
○	Area
○	Volume
○	Length

Jazz 19.3K/s 45% 2:31 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 202

Question # 13 of 30 (Start time: 02:31:49 PM, 29 December 2020)

Total Marks:

The unit normal to the surface $x + y - 2 = 0$ is

Select the correct option

Reload Math Equations

- $\frac{1}{\sqrt{2}}(\hat{k} + \hat{j})$
- $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$
- $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$
- does not exist

12:56

quiz.vu.edu.pk/QuizQu

MC200204967: KASHIF MEHMOOD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 12:56 PM, 29 December 202

Question # 1 of 30 (Start time: 12:56:36 PM, 29 December 2020)

Total Marks:

Evaluate $\int_c (e^{xyz})dy$ along the line segment joining $(0, 0, 0)$ to $(0, 1, 0)$.

Select the correct option

Reload Math Equations

- 1
- -1

Click to Save Answer & Move to Next Question

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MC190403973 LAILA SAIF

MTH622:Grand Quiz

Question # 24 of 30 (Start time: 11:42:13 AM, 29 December 2020)

Quiz Start Time: 11:07 AM, 29 December 2020

If φ and ψ are scalar point functions then $\nabla(\varphi\psi) =$

Total Marks: 1

Select the correct option

- $\varphi\nabla\psi - \psi\nabla\varphi$
- $\nabla\varphi \times \nabla\psi$
- $\nabla\varphi + \nabla\psi$
- $\varphi\nabla\psi + \psi\nabla\varphi$

quiz.vu.edu.pk/QuizQuestion.aspx?ver=b890b32b-18b1-49e3-b709-83...

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MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 03:26 PM,

Question # 4 of 30 (Start time: 03:28:14 PM, 29 December 2020)

Green's theorem implies that if $\oint_c Mdx + Ndy = 0$, then $\frac{\partial N}{\partial y} = \frac{\partial M}{\partial x}$.

Select the correct option

Reload

- true
- false

MC200205156: NIGHAT NOREEN

Time Left 78 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 13 of 30 (Start time: 04:04:39 PM, 29 December 2020)

Total Marks: 1

If $\nabla . \vec{V} = 0$ everywhere in some region of R, then $\vec{V}$ is called _______point function in the region.

Select the correct option

Reload Math Equations

- zero vector
- unit vector
- co-initial vector
- solenoid vector

Click to Save Answer & Move to Next Question

MTH622:Grand Quiz

Quiz St

Question # 22 of 30 (Start time: 03:48:56 PM, 29 December 2020)

When a particle moves along a straight path, then the particle has

Select the correct option

- tangential acceleration only
- centripetal acceleration only
- both tangential and centripetal acceleration
- none of these

MC200204967: KASHIF MEHMOOD

Time Left 67 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 12:56 PM, 29 December 202

Question # 4 of 30 (Start time: 12:59:20 PM, 29 December 2020)

Total Marks:

$\nabla . \vec{V}$ is a _______ quantity

Select the correct option

Reload Math Equations

- Scalar
- Vector

Click to Save Answer & Move to Next Question

Question # 1 of 30 (Start time: 12:50:47 PM, 29 December 2020)

If a particle of mass 2 units has position vector $\vec{r} = t^3\hat{i} + 8t\hat{j} - 9t\hat{k}$ then find the force acting on the particle.

Select the correct option

○	$12t$
○	$12t\hat{i}$

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MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 29 of 30 (Start time: 03:30:21 PM, 29 December 2020)

Total Marks: 1

A unit normal vector to a point of the positive side of a two sided surface S is called a _______ drawn unit normal.

Select the correct option

- Non-positive
- Negative
- Inward
- Outward

MC200204419: ABDUL HAFEEZ

Time Left 79 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:27 PM, 29 December 2020

Question # 25 of 30 (Start time: 02:53:57 PM, 29 December 2020)

Total Marks: 1

The directional derivative $\frac{d\varphi}{ds} = |\nabla\varphi||\hat{u}|\cos\theta$ is zero, when θ = ________

Select the correct option

Reload Math Equations

- 0°
- 180°
- 90°
- -180°

Quiz
quiz.vu.edu.pk

MC190203990: MUHAMMAD REHAN MURAD

Time Left 76 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 27 of 30 (Start time: 03:27:15 PM, 29 December 2020)

Total Marks:

Which of the following theorem convert line integral to surface integral?

Select the correct option

- Stoke's theorem only
- Gauss divergence and Stoke's theorem
- Green's theorem only
- Stoke's and Green's theorem

1:20

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2

MC200204967: KASHIF MEHMOOD

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 12:56 PM, 29 December 2020

Question # 28 of 30 (Start time: 01:20:00 PM, 29 December 2020)

Total Marks:

If ϕ and ψ are scalar point functions with continuous second order derivatives in a region R bounded by a closed surface S then Green's first identity states that

Select the correct option

Reload Math Equations

- $\int\int\int_R(\phi\nabla^2\psi + \nabla\phi\nabla\psi)dV = \int\int_S(\phi\nabla\psi).d\vec{S}$
- $\int\int\int_R(\phi\nabla^2\psi + \nabla\phi\nabla\psi)dV = \int\int_S(\psi\nabla\phi)d\vec{S}$

Click to Save Answer & Move to Next Question

Apps Gmail YouTube Maps Translate New Tab (6) What

MC190406995: AKHTAR HASSAN NOORI

MTH622:Grand Quiz

Quiz Start Time: 03:

Question # 21 of 30 (Start time: 03:47:32 PM, 29 December 2020)

Which of the following is NOT an application of Green's theorem?

Select the correct option

○	Centroid of plane figures
○	Volume of plane figures
○	Area surveying
●	Solving two dimensional flow integrals

Click to Save Answer & Mov

Quiz

MC200203833: FAIZAN HASSAN

MTH622:Grand Quiz

Time Left 88 sec(s)

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 26 of 30 (Start time: 04:05:33 PM, 29 December 2020)

Total Marks: 1

Evaluate $\int_0^1 \int_0^1 \int_0^1 x \, dx dy dz =$

Select the correct option

Reload Math Equations

○	$\frac{1}{2}$
○	$\frac{-1}{2}$

Click to Save Answer & Move to Next Question

Jazz 15.4K/s 4G 46% 2:30 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 11 of 30 (Start time: 02:30:48 PM, 29 December 2020)

Total Marks: 1

$\nabla\varphi.\hat{j} =$ ________.

Select the correct option

Reload Math Equations

○	$\frac{\partial\varphi}{\partial x}$
○	$\frac{\partial\varphi}{\partial y}$
○	$\frac{\partial\varphi}{\partial z}$
○	$\frac{\partial\varphi}{\partial y}\hat{j}$

Click to Save Answer & Move to Next Question

MTH622:Grand Quiz

Question # 1 of 30 (Start time: 11:58:01 AM, 29 December 2020)

According to first law of motion if some external force is acting on a body then the velocity of object is constant.

Select the correct option

- true
- false

Quiz
quiz.vu.edu.pk

MC190203990: MUHAMMAD REHAN MURAD

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 22 of 30 (Start time: 03:17:51 PM, 29 December 2020) Total Marks: 1

The value of $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ for the curve $x = t^2$, $y = 2t - 1$ and $z = 3t^2$ is

Select the correct option

Reload Math Equations

- $d\vec{r} = (2t + 2 + 6t)dt$
- $d\vec{r} = (2t\hat{i} + 2\hat{j} + 6t\hat{k})dt$
- $d\vec{r} = t\hat{i} + 2\hat{j} + 6t\hat{k}$
- $d\vec{r} = (t^2\hat{i} + 2t\hat{j} + 3t^2\hat{k})dt$

Click to Save Answer & Move to Next Question

MTH622:Grand Quiz

Question # 4 of 30 (Start time: 10:58:06 AM, 29 December 2020)

The identity $\oint_C \vec{B} \times d\vec{r} = \int\int_S (\hat{n} \times \vec{\nabla}) \times \vec{B} dS$ is

Select the correct option

○	true
○	false

Type here to search

3:09 pm 58%

Quiz
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MC190203990: MUHAMMAD REHAN MURAD

Time Left 79 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 13 of 30 (Start time: 03:09:11 PM, 29 December 2020)

Total Marks: 1

If $\vec{v} = y\hat{j}$, then the vector field $\vec{v}$ is

Select the correct option

Reload Math Equations

- solenoidal
- non solenoidal

Click to Save Answer & Move to Next Question

Question # 8 of 30 (Start time: 12:06:23 PM, 29 December 2020)

The value of $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ for the curve $x = t + 1,\ y = 2t^2$ and $z = t^3$ is

Select the correct option

- $d\vec{r} = (1 + 4t + 3t^2)dt$
- $d\vec{r} = (\hat{i} + t\hat{j} + t^2\hat{k})dt$
- $d\vec{r} = t\hat{i} + 4t\hat{j} + 3t^2\hat{k}$
- $d\vec{r} = (\hat{i} + 4t\hat{j} + 3t^2\hat{k})dt$

3:43 p.m. 47%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 6 of 30 (Start time: 03:43:28 PM, 29 December 2020)

Total Marks: 1

$\oint_C d\vec{r} \times \vec{B}$ for $\vec{B} = \hat{j}$ becomes

Select the correct option

Reload Math Equations

- $\oint_C (-dz\hat{i} - dx\hat{k})$
- $\oint_C (-dz\hat{i} + dx\hat{k})$

Click to Save Answer & Move to Next Question

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31

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 10:43 AM, 29 December 2020

Question # 6 of 30 (Start time: 10:48:31 AM, 29 December 2020)

Total Marks:

A curve which does not intersect itself anywhere is known as simple

Select the correct option

- Curve
- Path
- Closed curve
- None of above

Click to Save Answer & Move to Next Question

MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 14 of 30 (Start time: 03:10:05 PM, 29 December 2020)

Total Marks:

Gradient of a scalar function represents a ________ vector to the surface.

Select the correct option

- Tangent
- Secant
- Normal
- Unit

$$\int_0^1\int_0^1\int_0^1 \vec{A}\cdot dx\cdot dy\cdot dz \qquad \vec{A} = \hat{i}$$

$$= \int_0^1\int_0^1\int_0^1 \hat{i}\cdot dx\cdot dy\cdot dz$$

$$= \int_0^1\int_0^1 (x)\Big|_0^1 \hat{i}\cdot dy\cdot dz$$

$$= \int_0^1\int_0^1 (1-0)\hat{i}\cdot dy\cdot dz$$

$$= \int_0^1\int_0^1 \hat{i}\cdot dy\cdot dz = \int_0^1 (y)\Big|_0^1 \hat{i}\cdot dz$$

$$= \int_0^1 (1-0)\hat{i}\cdot dz = \int_0^1 \hat{i}\cdot dz$$

$$= (z)\Big|_0^1 \hat{i} = (1-0)\hat{i} = \hat{i}$$

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 2 of 30 (Start time: 03:39:01 PM, 29 December 2020)

Total Marks: 1

The functions in Green's theorem will be

Select the correct option

Reload Math Equations

- Continuous derivatives
- Discrete derivatives
- Continuous partial derivatives
- Discrete partial derivatives

Click to Save Answer & Move to Next Question

Question # 5 of 30 (Start time: 12:02:56 PM, 29 December 2020)

If a particle of mass 2 units has position vector $\vec{r} = 4t^2\hat{i} + 8t\hat{k}$ then find the rate of change of particle's momentum.

Select the correct option

- 16
- $16\hat{i}$

4:01 p.m. 44%

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 21 of 30 (Start time: 04:01:13 PM, 29 December 2020)

Total Marks: 1

$\hat{k} \times \vec{\nabla}\varphi$ for $\varphi = x$ becomes

Select the correct option

Reload Math Equations

- $-\hat{j}$
- $\hat{j}$

Click to Save Answer & Move to Next Question

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MC190403973: LAILA SAIF

MTH622:Grand Quiz

Question # 25 of 30 (Start time: 11:43:44 AM, 29 December 2020)

_______ is the example of scalar field.

Select the correct option

- Velocity
- Displacement
- Acceleration
- Pressure

$$\boxed{\oint_C d\vec{r} \times \vec{B} = \iint_S (\hat{n} \times \nabla) \times \vec{B}\, dS}$$

$$\oint_C \vec{A}.d\vec{r} = \iint_S (\nabla \times \vec{A}).\hat{n}dS$$

$= \vec{B} \times \vec{C}$ where $\vec{C}$ a constant vector is. Then

$$\oint_C (\vec{B} \times \vec{C}).d\vec{r} = \iint_S (\nabla \times (\vec{B} \times \vec{C})).\hat{n}dS$$

$$\oint_C d\vec{r}.(\vec{B} \times \vec{C}) = \iint_S [(\vec{C}.\nabla)\vec{B} - \vec{C}(\nabla.\vec{B})].\hat{n}dS$$

$$(d\vec{r} \times \vec{B}) = \iint_S [(\vec{C}.\nabla)\vec{B}].\hat{n}dS - \iint_S [\vec{C}(\nabla.\vec{B})].\hat{n}d$$

$$(d\vec{r} \times \vec{B}) = \iint_S \vec{C}.[\nabla(\vec{B}.\hat{n}]dS - \iint_S \vec{C}.[\hat{n}(\nabla.\vec{B})]dS$$

$$\vec{C}.\oint_C (d\vec{r} \times \vec{B}) = \vec{C}.\iint [\nabla(\vec{B}.\hat{n}) - \text{[illegible]}$$

$$\vec{C}.\oint (d\vec{r} \times \vec{B}) = \vec{C}.\iint \text{[illegible]}$$

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MC190401710: MALIK MUHAMMAD QASIM

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 27 of 30 (Start time: 02:43:49 PM, 29 December 2020)

Total Marks: 1

The normal component of acceleration is zero when motion of particle is

Select the correct option

- curvilinear
- rotational
- rectilinear
- translation

MC190403973: LAILA SAIF

MTH622:Grand Quiz

Question # 1 of 30 (Start time: 11:07:26 AM, 29 December 2020)

Newton's first law is also called

Select the correct option

- law of inertia
- law of conservat[illegible]entum

MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 20 of 30 (Start time: 03:16:15 PM, 29 December 2020)

Total Marks: 1

$\nabla\varphi = 0$ if and only if φ is

Select the correct option

Reload Math Equations

- Variable
- Undefined
- Constant
- Polynomial

Click to Save Answer & Move to Next Question

MTH622:Grand Quiz

Question # 10 of 30 (Start time: 11:03:58 AM, 29 December 2020)

If the vector $\vec{V} = bx\hat{i} + (y-1)\hat{j}$ is solenoidal, then the value of constant b is

Select the correct option

○	-1
○	0
○	±1
○	1

Type here to search

Quiz

MC200203833: FAIZAN HASSAN

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:37 PM, 29 December 2020

Question # 24 of 30 (Start time: 04:03:44 PM, 29 December 2020)

Total Marks: 1

The operator ∇^2 is defined by

Select the correct option

Reload Math Equations

- $\nabla^2 = \frac{\partial^2}{\partial x^2}\hat{i} + \frac{\partial^2}{\partial y^2}\hat{j} + \frac{\partial^2}{\partial z^2}\hat{k}$
- $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

Click to Save Answer & Move to Next Question

Jazz 15.7K/s 4G 43% 2:34 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 16 of 30 (Start time: 02:34:32 PM, 29 December 2020)

Total Marks: 1

The Green's theorem in plane can be written in tangential form as

Select the correct option

Reload Math Equations

- $\oint_C \vec{B}.\hat{n}dS = \int\int_S \vec{\nabla}.\vec{A}dR$
- $\oint_C \vec{A}.d\vec{r} = \int\int_R (\vec{\nabla}\times\vec{A}).\hat{k}dR$

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Quiz

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MC200205156: NIGHAT NOREEN

Time Left 83 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 7 of 30 (Start time: 03:58:55 PM, 29 December 2020) Total Marks: 1

The parametric form of the straight line joining (0, 0, 0) and (1, 1, 1) is

Select the correct option

Reload Math Equations

- $x = t,\ y = 2t$ and $z = t^2$
- $x = 1,\ y = 1$ and $z = 1$
- $x = t,\ y = t$ and $z = t$
- $x = t,\ y = t^2$ and $z = t^3$

Click to Save Answer & Move to Next Question

Question # 16 of 30 (Start time: 12:41:55 PM, 29 December 2020) Total Marks: 1

A unit normal vector to any point of the positive side of a two sided surface S is called a _______ drawn unit normal.

Select the correct option

- Positive
- Negative
- Inward
- None of these

Click to Save Answer & Move to Next Question

Jazz 13.9K/s 4G 43% 2:33 PM

quiz.vu.edu.pk/QuizQue 22

MC190401710: MALIK MUHAMMAD QASIM

Time Left 89 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:22 PM, 29 December 2020

Question # 14 of 30 (Start time: 02:33:12 PM, 29 December 2020)

Total Marks: 1

A simple closed curve does not intersects itself anywhere.

Select the correct option

Reload Math Equations

○	true
○	false

$\vec{v} = i$

$\nabla \cdot v = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot i$

$\nabla \cdot v = 0$

$\therefore$ v is solenoidal

15:58 67%

Quiz

https://quiz.vu.edu.pk/QuizQuestio...

MC200205156: NIGHAT NOREEN

Time Left 87 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 03:53 PM, 29 December 2020

Question # 6 of 30 (Start time: 03:58:08 PM, 29 December 2020)

Total Marks: 1

Which of the following theorem convert line integral to surface integral?

Select the correct option

- Stoke's theorem only
- Gauss divergence and Stoke's theorem
- Green's theorem only
- Stoke's and Green's theorem

MC190203990: MUHAMMAD REHAN MURAD

Time Left 86 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 6 of 30 (Start time: 02:57:56 PM, 29 December 2020)

Total Marks: 1

Evaluate $\int_0^1 \int_0^1 \int_0^1 x \, dx dy dz =$

Select the correct option

Reload Math Equations

- $\frac{1}{2}$
- $-\frac{1}{2}$

○	-1
○	0
○	1
○	3

Question # 17 of 30 (Start time: 12:42:35 PM, 29 December 2020) Total Marks: 1

$\nabla \times \vec{A}$ is used to denote

Select the correct option

Reload Math Equations

a. Curl $\vec{A}$

b. Rot $\vec{A}$

c. Gradient $\vec{A}$

d. Both a and b

Quiz List

Quiz

z.vu.edu.pk/QuizQuestion.aspx?ver=9fbe61e0-4483-4174-be08-806...

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AR HASSAN NOORI

Time Left 86 sec(s)

Quiz Start Time: 03:26 PM, 29 December 2020

art time: 03:38:23 PM, 29 December 2020)

Total Marks: 1

$$\nabla\varphi.\,\hat{j} = \_\_\_\_\_\_.$$

on

Reload Math Equations

$$\frac{\partial\varphi}{\partial x}$$

$$\frac{\partial\varphi}{\partial y}$$

$$\frac{\partial\varphi}{\partial z}$$

$$\frac{\partial\varphi}{\partial y}\hat{j}$$

MC190203990: MUHAMMAD REHAN MURAD

Time Left 88 sec(s)

MTH622:Grand Quiz

Quiz Start Time: 02:51 PM, 29 December 2020

Question # 9 of 30 (Start time: 03:01:52 PM, 29 December 2020)

Total Marks: 1

If the vector $\vec{V} = 2x\hat{i} + az\hat{k}$ is solenoidal, then the value of constant a is

Select the correct option

Reload Math Equations

- 2
A
- 0
- -2
- 1

Question # 3 of 30 (Start time: 10:57:16 AM, 29 December 2020)

The Stoke's theorem uses which of the following operation?

Select the correct option

- Divergence
- Gradient
- Curl
- Laplacian

Question # 29 of 30 (Start time: 12:51:48 PM, 29 December 2020) Total Marks: 1

Divergence is ______ operator.

Select the correct option

- scalar
- vector
- directional
- None of these